

# Fabrizio **Miano**


Business Intelligence · Data Solutions · Cloud Computing

*Catania, Italy*

(+39) 333 60 66 844 | fabriziomiano@gmail.com | fabriziomiano.com | fabriziomiano | fabriziomiano


## Overview

Business Analytics Manager at STMicroelectronics, developing data solutions and analytical tools for Sales Operations and turning complex datasets into actionable decisions across microelectronics and supply-chain environments. Certified SAFe® 5 Scrum Master with experience in business analytics, solution design, project delivery, and cross-functional collaboration.

Previously led data-quality and governance initiatives at Coca-Cola HBC and international agile analytics projects for STMicroelectronics at Capgemini. PhD in Experimental Particle Physics from the University of Sussex and CERN as part of the ATLAS experiment; combines rigorous data analysis with clear communication, including national and international presentations and scientific outreach.

## Industrial Experience

### STMicroelectronics

*Catania, Italy*

Business Analytics Manager — *Jan 2025 – Present*

- Leading business analytics activities supporting Sales Operations and data-driven decision-making;
- Designing and implementing robust data solutions for microelectronics and supply-chain dynamics;
- Coordinating with cross-functional stakeholders to translate business needs into analytical tools and decision-support assets;

### STMicroelectronics

*Catania, Italy*

Senior Business Advisor / Business Operations Support PM — *Jun 2024 – Jan 2025*

- Designing and maintaining analytical tools and dashboards for KPI tracking;
- Utilizing scripting and programming skills to automate tasks and support Sales, Marketing, and Operations teams;
- Managing small data projects to improve operational efficiency and support strategic initiatives;

### Coca-Cola HBC

*Remote*

Senior DQ Analyst — *May 2023 – May 2024*

- Responsible for implementing and executing the business-led data governance initiatives to hold tight on the quality of the upstream part of the data value chain;
- translation of the data-quality business requirements to technical solutions;
- hands-on: implementation of the data quality detection and remediation through automation and modern data techniques;
- embrace agile: understand the iterative nature of data projects and welcome the changes to ensure high satisfaction for the downstream data users, and collaborate with multiple stakeholders and data users, including data curator, data engineers, data scientists, functional teams and functional data owners

### Capgemini

*Remote*

Senior Consultant — *Mar 2021 – Apr 2023*

Senior Data Engineer and Team Leader in an international agile project focused on the delivery of an end-to-end business intelligence solution to explore customer demand and demand planning forecast analytics, fully developed on the Azure stack: Databricks, Data Factory, SQL Database, Synapse Analytics, Analysis Services, and PowerBI; DevOps and versioning best-practice with Jenkins and Terraform; code reviewer on Gerrit/GitHub/Azure Repos

### Experis IT

*Remote*

Business Intelligence Consultant — *Sep 2020 – Mar 2021*

Data-engineering consultancy for Capgemini with MS Azure

### R2M Solution

*Remote*

ICT Specialist / PM — *Jan 2020 – Sep 2020*

- Full-stack development of a custom ERP (Frappé)
- System administrator for on-premises and cloud-based (Google cloud, AWS, Azure, and Heroku) web solutions written in Python and Bootstrap
- Technical activities in research projects within the robotics, artificial intelligence, big data, energy saving, and IoT domains
- Delivery of high quality reports for research and innovation projects
- IT Service Management and help desk

### Prisma

*Catania, Italy*

Data Scientist / Backend Developer

*Jan 2019 – Jan 2020*

- Data crawling: development of a crawler written in Python to get jobs and user profile pages from social networks
- Recommendation system: development of a ML tool (Python) to perform job-user matching in order to recommend the most suitable job for a given user
- Computer Vision: development of the back-end of a Web Application (Python, Flask) to perform real-time object detection using YOLO
- NLP: development of a deep-learning tool to extract information from a PDF; RESTful API integration (Python, Flask)
- Social-network Analysis: named-entity extraction and sentiment analysis on Facebook-post comments; RESTful API (Python, Flask)
- Management of micro-services on Azure with Docker and Azure Container Registry

### Deckchair.com

*Brighton, UK*

Junior Data Scientist

*Jun – Aug 2018*

- Development of a statistical tool to select interesting images within a dataset (Python, skimage);
- Design of a ML tool to classify grey sky and sunset images (Python, sklearn, Keras);

## Academic Experience

### University of Sussex

*Brighton, UK*

Post-Doctoral Research Fellow in Data Science

*Mar – Dec 2018*

- Development of ML model to classify videos of trials as Truthful/Deceptive (Python, Keras, Tensorflow)
- Exploration of potential collaboration with other areas of research such as, Evolutionary and Adaptive Systems Research Group, Industrial IT and Signal Processing Research Group, Sensor Technology Research Centre

### University of Sussex / CERN

*Brighton, UK / Geneva, Switzerland*

Doctoral Position

*Nov 2014 – Oct 2018*

- Developed Python code to perform multi-variable signal-over-background optimisations to define signal-enriched regions to enhance signal contribution while discriminating against background
- Modelling of main backgrounds using custom C++ framework
- Data-Driven estimation of an *irreducible* background and evaluation of related theory uncertainties (C++, Python, Excel)
- Participation in the development and commissioning of vertex reconstruction algorithm (C++)
- Validation of the changes to the used framework before release
- Monitoring of the performance of the tracking efficiency
- Development of a monitoring tool (Python) to display the results of the monitoring (plots, tables) on a web page (HTML)

### Università degli Studi di Catania

*Catania, Italy*

MSc INFN-funded project

*Mar 2013 – Jul 2014*

- Developed C++ code to simulate physical distributions of the emission of protons from a nuclear source of a given size and life-time
- Validation of the simulations
- Deduction of the geometrical size and life-time of a nuclear source created in a heavy-ion collision at intermediate energy using data collected with the LASSA detector of the NSCL, Michigan, US.

## Education

### University of Sussex / CERN

*Brighton, UK / Geneva, Switzerland*

PhD in Experimental Particle Physics

*2015–2018*

Optimisation studies and data-driven background estimation in searches for the supersymmetric partner of the top quark with the ATLAS Detector at the LHC

### Università degli Studi di Catania

*Catania, Italy*

MSc in Experimental Nuclear Physics

*2009–2014*

Protons emission and space-time characterisation in heavy-ion collisions

### Università degli Studi di Catania

*Catania, Italy*

BSc in Physics

*2005–2009*

Study of neutrons spectra emitted by Am-Be and Pu-Be radioactive sources

## Tech Skills

| | |
|---|---|
| **Languages** | Python, SQL, Bash, HTML, JavaScript, C/C++, LaTeX |
| **Database** | MongoDB, MySQL, SQL Server |
| **Versioning** | Git, SVN |
| **Web Servers** | NGINX, Apache |
| **Azure** | SQL Server, Databricks, Data Factory, Synapse Analytics, Analysis Services, App Service, Container instances, Functions, VMs, CosmosDB, EventHub |
| **AWS** | EC2, Elastic Beanstalk, Lambda |
| **Heroku** | Apps and Pipelines |
| **Data Science** | sklearn, skimage, Keras, pandas, numpy, OpenCV, spaCy, NLTK, librosa, selenium, BeautifulSoup |
| **DevOps** | Docker, Jenkins, Terraform |
| **OSs** | Linux, macOS, Windows |

## Soft Skills

| | |
|---|---|
| **Teaching** | Data-science group mentor; Classical Physics and Electromagnetism; Properties of Matter |
| **Languages** | English (Fluent), Spanish (Fluent), Italian (Mother tongue), French (Basic) |

## Public Speaking

**Wider Topics in Data Science seminar series** — *University of Sussex, Brighton, UK*

"5 years later: story of a transition from a PhD in Physics to a career in data science" — *Mar 2023*

1-hour seminar on the first five working years after the PhD

**Wider Topics in Data Science seminar series** — *University of Sussex, Brighton, UK*

"From academia to industry" — *Feb 2021*

1-hour seminar on the transition from academia to industry

**DISCnet showcase** — *Royal Society, London, UK*

"From Particle Physics to Computer Vision" — *Dec 2018*

10-min talk on the ML tool developed for Deckchair.com

**HLT-UK Internal Meeting** — *University of Oxford, UK*

"HLT tracking performance" — *Sep 2017*

15-min talk: Presented the result of the performance of the tracking of the ATLAS Inner detector

**Phenomenology 2017** — *Pittsburgh, PA, USA*

"Searches for direct production of third generation squarks with the ATLAS detector" — *May 2017*

15-min talk: Presented the results of the searches for third-generation squarks on behalf of the ATLAS Collaboration

**International Conference on High Energy Physics (ICHEP) 2016** — *Chicago, IL, USA*

"The Design and performance of the ATLAS Inner Detector Trigger for Run 2 collisions at $\sqrt{s} = 13$ TeV" — *Aug 2016*

Poster presented on behalf of the ATLAS Collaboration: Results of the performance of the ATLAS Inner Detector

**STFC HEP Summer School** — *Lancaster, UK*

"Direct pair production of the top squark in all-hadronic final states in *pp* collisions with the ATLAS detector" — *2015*

Poster: preliminary results of the optimisation of the regions of interest for the search of the supersymmetric partner of the top quark

## Awards

| | | |
|---|---|---|
| 2018 | **Post-doctoral Fellowship,** STFC Impact Acceleration Account | *Brighton, UK* |
| 2018 | **Grant,** DISCnet | *Brighton, UK* |
| 2016 | **Doctoral grant,** Doctoral Overseas Conference Grant for Postgraduate Researchers | *Chicago, IL, USA* |
| 2015 | **Scholarship,** 4-year PhD scholarship in collaboration with CERN | *CERN, Switzerland* |
| 2013 | **Scholarship,** 1-year MSc scholarship in collaboration with LNS | *Catania, Italy* |

## Outreach

| | | |
|---|---|---|
| Jun 2018 | **#ScienceOnBuses,** Participated in filming and engaged people on buses about CERN research activities | *Brighton, UK* |
| Nov 2017 | **HiSPARC,** Helped middle school student build a scintillator for cosmic-ray detection | *Brighton, UK* |
| May 2016 | **CERN Master class,** Talked to a high-school students audience about "The life of a PhD student at CERN" | *CERN, Switzerland* |
| Feb 2015 | **Brighton Science Festival,** Engaged kids through games related to particle physics | *Brighton, UK* |

# Publications

## International peer-reviewed conferences/proceedings

[6] F. Miano for the ATLAS Collaboration. “The design and performance of the ATLAS Inner Detector trigger for Run 2 LHC collisions at $\sqrt{s} = 13$ TeV”. In: *Proceedings, 38th International Conference on High Energy Physics (ICHEP 2016): Chicago, IL, USA, August 3-10, 2016*.

## PhD Thesis

[2] Fabrizio Miano and Fabrizio Salvatore. “Optimisation studies and data-driven background estimation in searches for the supersymmetric partner of the top quark with the ATLAS Detector at the LHC”. Presented 28 Sep 2018. Sept. 2018. URL: http://cds.cern.ch/record/2650559.

## Articles in peer-reviewed journals

[3] The ATLAS Collaboration. “Performance of the ATLAS trigger system in 2015”. In: *The European Physical Journal C* 77.5 (May 2017), p. 317. ISSN: 1434-6052. DOI: 10.1140/epjc/s10052-017-4852-3. URL: https://doi.org/10.1140/epjc/s10052-017-4852-3.

[4] The ATLAS Collaboration. “Search for a scalar partner of the top quark in the jets plus missing transverse momentum final state at $\sqrt{s}$=13 TeV with the ATLAS detector”. In: *JHEP* 12 (2017), p. 085. DOI: 10.1007/JHEP12(2017)085. arXiv: 1709.04183 [hep-ex].

[5] The ATLAS Collaboration. “Search for dark matter produced in association with bottom or top quarks in $\sqrt{s}$ = 13 TeV pp collisions with the ATLAS detector”. In: *The European Physical Journal C* ICHEP2016 (2016), p. 856. eprint: 1710.11412. URL: https://doi.org/10.1140/epjc/s10052-017-5486-1.

[6] The ATLAS Collaboration. “Search for the Supersymmetric Partner of the Top Quark in the Jets+$E_{\mathrm{T}}^{\mathrm{miss}}$ Final State at $\sqrt{s} = 13$ TeV”. In: *CERN Document Server* (2016).